
MODEL SELECTION AND VALIDATION REPORT (fishing_tric model)

OBSERVATIONS:1739

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (fishing_tric model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: gaussian(link = 'identity')

Conditional model:

	coef.est	coef.se
(Intercept)	-0.08	0.47
log_sum_fishing	0.00	0.02
sbt_mean	0.31	0.09
sbt_sd	0.02	0.04
sst_sd	-0.08	0.04
sic_mean	0.03	0.06
log_chla_mean	0.01	0.03
depth	-0.05	0.04

Dispersion parameter: 0.73

Spatiotemporal AR1 correlation (rho): 0.97

Matérn range: 444.31

Spatiotemporal marginal AR1 SD: 1.08

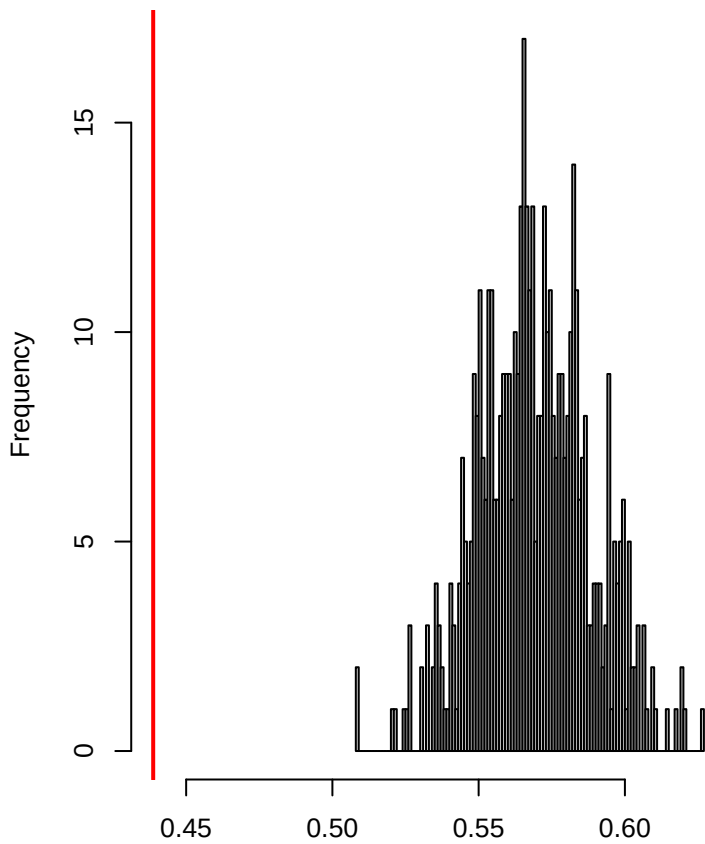
REML criterion at convergence: 2147.927

See ?tidy.sdmTMB to extract these values as a data frame.

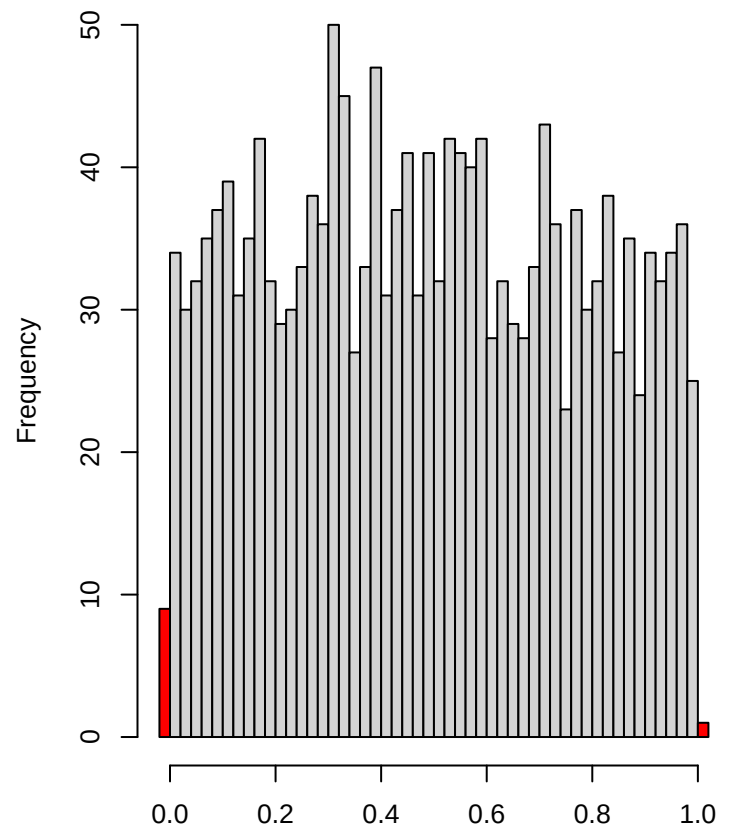
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	12	4319.854	1.2855152
spatiotemporal_random_fields_ar1_with_anisotropy	14	4321.139	180.6071231
spatial_random_fields	11	4501.747	0.0547766
spatial_random_fields_with_anisotropy	13	4501.801	46.4990787
spatiotemporal_random_fields_iid	11	4548.300	3.4986063
spatiotemporal_random_fields_iid_with_anisotropy	13	4551.799	286.0401780
error_only	9	4837.839	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



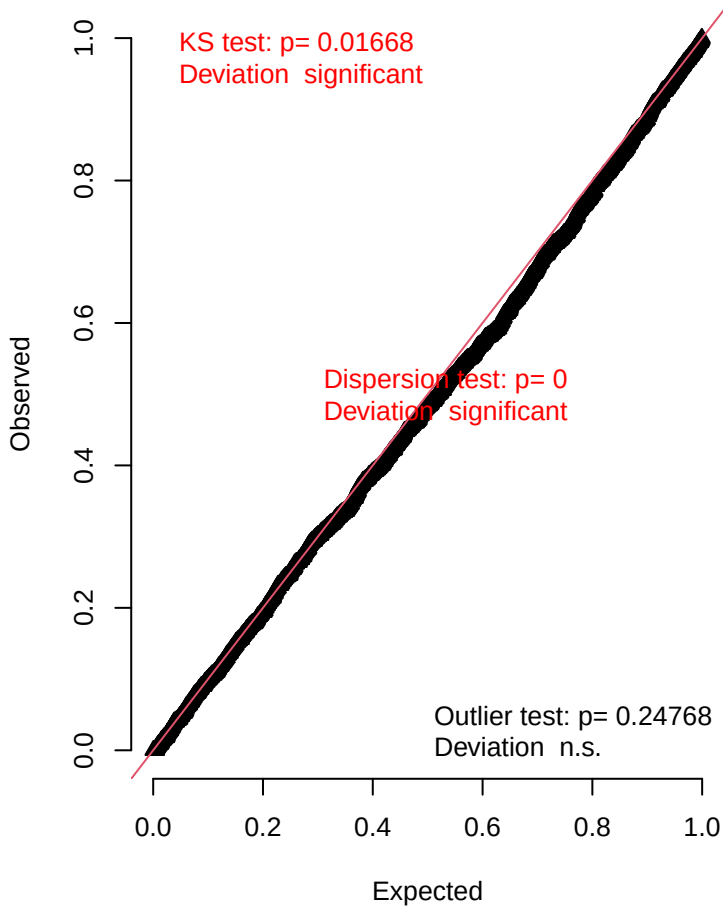
Outlier test n.s.



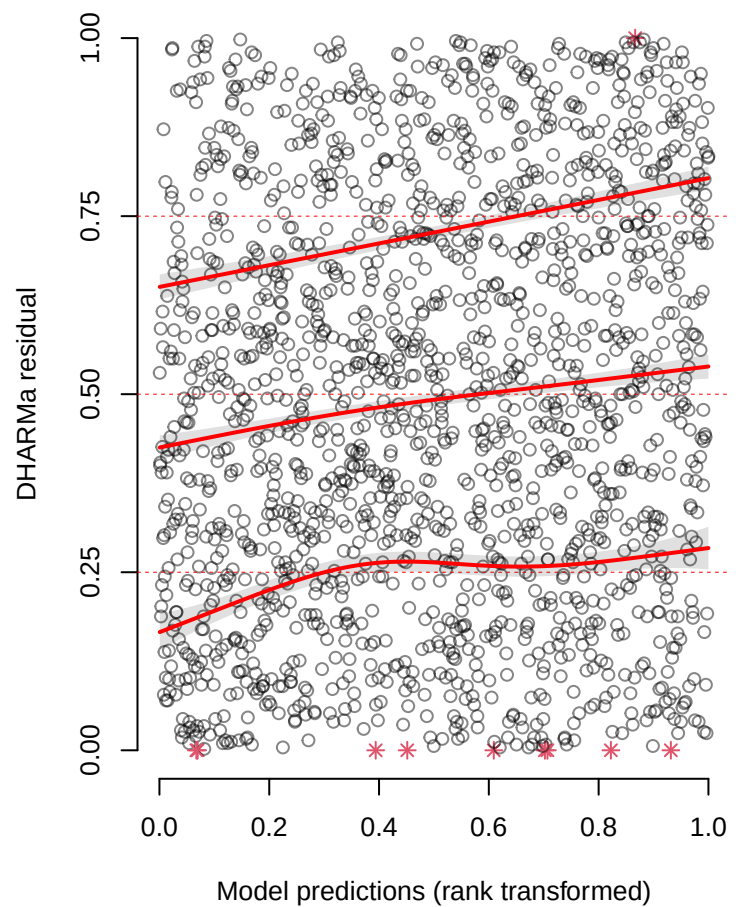
Simulated values, red line = fitted model. p-value (two.sided)

Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMA Moran's I test for distance-based autocorrelation
Moran's I = 0.023 | H1: Distance-based autocorrelation | p-value = 0.01

